

The axioms of n -exact categories

BACHELOR'S THESIS

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Introduction

Homological algebra is a key aspect of many modern topics in mathematics, be it geometry, algebra or topology. While originally only studied on modules and abelian groups, it was then later formalized on abelian categories, giving a more abstract framework. Exact categories, first introduced by Quillen, are a continuation of this abstraction, reducing the requirements to the bare minimum while still enabling discussing short exact sequences. In an attempt to also study longer exactness and to axiomatize properties of so-called n -cluster tilting subcategories of exact categories, a generalization to n -exact categories was introduced by Jasso. This thesis aims to build the necessary tools to study such structures, while also providing a few examples.

First, we will generalize the structure of short exact sequences to additive categories, motivated by fundamental properties arising in abelian categories. We will then proceed with some examples of exact categories by finding extension-closed subcategories. Finally, we will give a smaller list of axioms that is equivalent to the classical definition.

Then, we will extend the concepts studied to longer exact sequences. We will define the necessary higher-degree concepts in order to define n -exact categories, and proceed with a few elementary results. To conclude, we will give some examples of n -exact categories by looking at n -cluster tilting subcategories.

This thesis does not have a direct connection to sustainability. However, it is worth noting that quality education is one of the United Nations' Sustainable Development Goals. In particular, abstract algebra—and mathematics as a whole—fosters critical and rational thinking, which are essential skills for addressing complex sustainability challenges.

In order to understand this thesis, the reader should have some knowledge about modern homological algebra. In particular, they should

already be familiar with category theory, more specifically abelian category theory, as is presented for example in the lecture notes of MA3204 Homological Algebra [\[8\]](#).

For the last part of the thesis, the reader should also be familiar with Auslander-Reiten theory and AR-quivers, as is for example explained in [\[1\]](#).

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Conventions and notations

Throughout this thesis, we let n be a positive integer. We usually let $\mathcal{A}$ denote an additive category. When not stated otherwise, the word kernel (respectively cokernel) will refer to the kernel morphism (respectively cokernel morphism) instead of the object. For a morphism $f : A \rightarrow B$, we call A its domain and B its codomain. For two objects $A, B \in \mathcal{A}$, we let $\mathcal{A}(A, B)$ denote the set of morphisms from A to B . We will use $\ker f$ to refer to its kernel morphism and $\operatorname{cok} f$ for its cokernel morphism, whereas $\operatorname{Ker} f$ will refer to its kernel object and $\operatorname{Cok} f$ its cokernel object. For any object $A \in \mathcal{A}$, we denote by 1_A the identity morphism on A . We let $\operatorname{Ch}^n(\mathcal{A})$ denote the set of chain complexes on $\mathcal{A}$ concentrated in degrees $\{0, \dots, n\}$ and $\operatorname{Ch}(\mathcal{A})$ denotes the set of all chain complexes on $\mathcal{A}$. We let $\operatorname{H}(\mathcal{A})$ denote the homotopy category of $\mathcal{A}$.

1 An introduction to exact categories

As a way to generalize the notion of abelian categories, one can look at a general additive category on which we define a certain notion of short exact sequences that has very similar properties to the abelian category counterpart. This chapter aims to motivate the definition of these so-called exact categories, to later define them formally and give a few concrete examples. We will also give a minimal version of the axioms due to B. Keller.

The content of this chapter is inspired by the lecture notes of MA3204 Homological Algebra [8], the article about exact categories by T. Bühler [3] and B. Keller's notes on exact categories [5, Appendix A].

1.1 Motivation

When studying abelian categories, there are many convenient properties related to monomorphisms and epimorphisms, which are used to prove results about short exact sequences. Let us recall a few:

Proposition 1.1 *In an abelian category,*

- *every monomorphism is a kernel of its cokernel;*
- *every epimorphism is a cokernel of its kernel.*

Furthermore, the following properties hold:

- *the pushout of a monomorphism is a monomorphism;*
- *the pullback of a monomorphism is a monomorphism;*
- *the pushout of an epimorphism is an epimorphism;*
- *the pullback of an epimorphism is an epimorphism.*

We refer the reader to [8, Sections 12-13] for the proofs of these facts.

Fundamentally, this tells us that every monomorphism and every epimorphism fits into a short exact sequence, and that taking the pullback or pushout of a morphism that is part of a short exact sequence gives rise to another short exact sequence. Indeed, if $f : A \rightarrowtail B$ is a monomorphism, then it fits into the short exact sequence $A \xrightarrow{f} B \xrightarrow{\text{cok}f} \text{Cok } f$. Furthermore, since morphisms that are part of a short exact sequence are either monomorphisms or epimorphisms, they will once again be a monomorphism or an epimorphism after a pullback or pushout, thus will once again fit in a short exact sequence.

Let us quickly recall what it means for a class of composable morphisms to be closed under isomorphisms.

Definition 1.2 Let $\mathcal{E}$ be a class of composable morphisms, i.e. a set of pairs (f, g) where the codomain of f is the domain of g . We say $\mathcal{E}$ is *closed under isomorphisms* if, for any commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ a \downarrow & & b \downarrow & & \downarrow c \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array}$$

where $(f, g) \in \mathcal{E}$ and the vertical maps are isomorphisms, we have $(f', g') \in \mathcal{E}$.

There are many properties we can conclude about kernels and cokernels using the previous results and this definition. These properties are what will motivate the definition of exact categories.

Proposition 1.3 *Let $\mathcal{A}$ be an abelian category and let $\mathcal{E}$ be the class of short exact sequences in $\mathcal{A}$. Then $\mathcal{E}$ is closed under isomorphisms. Furthermore, the following hold:*

- [E0] *For any object $A \in \mathcal{A}$, the identity morphism 1_A is a kernel;*
- [E0^{op}] *For any object $A \in \mathcal{A}$, the identity morphism 1_A is a cokernel;*

- [E1] *The class of kernels is closed under composition;*
- [E1^{op}] *The class of cokernels is closed under composition;*
- [E2] *The pushout of a kernel along any morphism is a kernel;*
- [E2^{op}] *The pullback of a cokernel along any morphism is a cokernel.*

Proof It is clear that the class of short exact sequences is closed under isomorphisms. Indeed, $A \xrightarrow{f} B \xrightarrow{g} C$ is a short exact sequence if and only if f is a kernel of g and g is a cokernel of f . The result follows from the fact that kernels and cokernels are preserved by isomorphisms.

[E0] and [E0^{op}] follow directly from the fact that the identity morphism is both the kernel and cokernel of the zero map.

We can deduce [E1] and [E1^{op}] by recalling that composing monomorphisms (reps. epimorphisms) gives us monomorphisms (resp. epimorphisms) and that monomorphisms are kernels of their cokernel (resp. epimorphisms are cokernels of their kernel).

As for [E2], recall that kernels are always monomorphisms. Thus, by theorem 1.1, the pushout of a kernel is a monomorphism as well. But then, by theorem 1.1 again, it is a kernel of its cokernel, thus it is a kernel. The proof for [E2^{op}] is similar. $\square$

Remark 1.5 We can in a very similar way prove that the pullback of a kernel (respectively the pushout of a cokernel) is a kernel (respectively cokernel). However, we choose not to include this as it turns out to be unnecessary for many results.

1.2 Exact categories

Based on the observations for abelian categories from the last section, one can define a more general notion that encapsulates the idea of short exact sequences, namely exact categories. To do so, let us first define kernel-cokernel pairs.

Definition 1.6 Let $\mathcal{A}$ be an additive category. A *kernel-cokernel pair* (i, p) in $\mathcal{A}$ is a pair of composable morphisms $A \xrightarrow{i} B \xrightarrow{p} C$ such that i is a kernel of p and p is a cokernel of i .

In the context of abelian categories, we notice that kernel-cokernel pairs indeed give rise to short exact sequences in the usual sense. In a general additive category, however, we cannot say the same since we have no guarantee that kernels and cokernels even exist for arbitrary morphisms, and thus images need not exist. But we will use the intuition developed in the context of abelian categories to generalize the concept of short exact sequences to the setting of additive categories. First, let us fix some terminology.

Definition 1.7 Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$. We call a morphism $i : A \rightarrow B$ an *inflation* (or *admissible monomorphism*) if there exists a morphism $p : B \rightarrow C$ such that $(i, p) \in \mathcal{E}$. Dually, we call a morphism $q : B \rightarrow C$ a *deflation* (or *admissible epimorphism*) if there exists a morphism $j : A \rightarrow B$ such that $(j, q) \in \mathcal{E}$. We use the notation $A \rightarrowtail B$ for inflations, and $B \twoheadrightarrow C$ for deflations. We say that $\mathcal{E}$ is *closed under isomorphisms* if whenever we have $A \rightarrowtail B \twoheadrightarrow C \in \mathcal{E}$ fitting in a commutative diagram

$$\begin{array}{ccccc} A & \rightarrowtail & B & \xrightarrow{p} & C \\ a \downarrow & & b \downarrow & & \downarrow c \\ A' & \xrightarrow{j} & B' & \xrightarrow{q} & C' \end{array}$$

with a , b , and c isomorphisms, then $(j, q) \in \mathcal{E}$.

It is usually convenient to require being closed under isomorphisms, as objects with universal properties are generally unique up to isomorphism. In particular, recall that all kernel objects of a specific morphism are isomorphic. In the case of a class of kernel-cokernel pairs $\mathcal{E}$, being closed under isomorphisms implies that all kernels of a deflation are inflations. Indeed, consider $p : B \twoheadrightarrow C$ a deflation and $i : A \rightarrowtail B$

an inflation such that $(i, p) \in \mathcal{E}$. Let $j : A' \rightarrow B$ be another kernel of p and $a : A \rightarrow A'$ be the isomorphism between both kernels. The sequence $A' \xrightarrow{j} B \xrightarrow{p} C$ fits in the commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{p} & C \\ a \downarrow & & \parallel & & \parallel \\ A' & \xrightarrow{j} & B & \xrightarrow{p} & C \end{array}$$

Since all vertical arrows are isomorphisms, we conclude that $(j, p) \in \mathcal{E}$, so j is an inflation. The proof that all cokernels of an inflation are deflations is dual.

We can now proceed to define exact categories, basing our definition on the properties we established for abelian categories.

Definition 1.8 Let $\mathcal{A}$ be an additive category. An *exact structure* on $\mathcal{A}$ is a class $\mathcal{E}$ of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphisms and satisfies the following axioms:

- [E0] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an inflation;
- [E0^{op}] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is a deflation;
- [E1] The composition of two inflations yields an inflation;
- [E1^{op}] The composition of two deflations yields a deflation;
- [E2] The pushout of an inflation along any morphism yields an inflation;
- [E2^{op}] The pullback of a deflation along any morphism yields a deflation.

The last two axioms can be summarized by the following commutative diagrams:

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ c \downarrow & \lrcorner & \downarrow d \\ C' & \xrightarrow{f'} & D' \end{array} \quad \text{and} \quad \begin{array}{ccc} C' & \dashrightarrow^{f'} & D' \\ \downarrow c & \lrcorner & \downarrow d \\ C & \xrightarrow{f} & D \end{array}$$

If $\mathcal{E}$ is an exact structure on $\mathcal{A}$, we say that $(\mathcal{A}, \mathcal{E})$ is an *exact category* and the elements of $\mathcal{E}$ are called *conflations* (or *short exact sequences*).

1.3 Extension-closed subcategories

It follows immediately from section 1.1 that all abelian categories are exact categories, with a possible exact structure being all short exact sequences. However, one may also look at specific subcategories of an abelian category, namely full extension-closed subcategories, to get examples of non-abelian exact categories. Let us first clarify what this means.

Definition 1.9 Let $\mathcal{A}$ be an abelian category and $\mathcal{C} \subseteq \mathcal{A}$ be a full additive subcategory of $\mathcal{A}$. $\mathcal{C}$ is said to be *closed under extension* if, for any short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

in $\mathcal{A}$ such that A and C are both in $\mathcal{C}$, then B is also in $\mathcal{C}$.

In that context, we say that B is an *extension of C by A* .

As mentioned previously, this is the only requirement for a full subcategory of an abelian category to be exact, with the exact structure inherited from the short exact sequences in $\mathcal{A}$.

Proposition 1.10 *Let $\mathcal{A}$ be an abelian category and $\mathcal{C} \subseteq \mathcal{A}$ be a full additive subcategory of $\mathcal{A}$ that is closed under extensions. Let $\mathcal{E}$ be the collection of short exact sequences in $\mathcal{A}$ such that all three objects are in $\mathcal{C}$. Then $(\mathcal{C}, \mathcal{E})$ is an exact category.*

Proof We prove each of the axioms one by one.

Note that $\mathcal{E}$ is closed under isomorphisms since short exact sequences in $\mathcal{A}$ are closed under isomorphisms.

Since $\mathcal{C}$ is an additive subcategory, it contains the zero object. As such, for any object $B \in \mathcal{C}$, the sequences $0 \rightarrow B \xrightarrow{1_B} B$ and $B \xrightarrow{1_B} B \rightarrow 0$ are short exact and contained in $\mathcal{C}$, i.e. identity morphisms are both deflations and inflations. Thus, both $[\text{E0}]$ and $[\text{E0}^{\text{op}}]$ are satisfied.

Now let us consider the axiom $[\text{E2}]$. Let $X \xrightarrow{i} Y \xrightarrow{d} Z$ be a short exact sequence in $\mathcal{C}$ and $f : X \rightarrow A$. We can consider the pushout diagram

$$\begin{array}{ccccc} X & \xrightarrow{i} & Y & \xrightarrow{d} & Z \\ f \downarrow & & f' \downarrow & & f'' \downarrow \\ A & \xrightarrow{j} & B & \xrightarrow{e} & C \end{array}$$

in the ambient category $\mathcal{A}$, where $C = \text{Cok } j$. Recall, from theorem 1.1, that j is a monomorphism. Furthermore, it is well known that we can choose $C = Z$; see [8, Proposition 13.7]. Thus the sequence $A \xrightarrow{j} B \xrightarrow{e} Z$ is short exact in $\mathcal{A}$, with both A and Z in $\mathcal{C}$, which implies that $B \in \mathcal{C}$ since $\mathcal{C}$ is extension-closed. We therefore get a short exact sequence in $\mathcal{C}$, and j is an inflation. The proof for $[\text{E2}^{\text{op}}]$ is dual.

To prove $[\text{E1}]$, let us consider short exact sequences $A \xrightarrow{i} B \xrightarrow{d} C$ and $B \xrightarrow{j} Y \xrightarrow{e} Z$ in $\mathcal{C}$. Then we can construct the following pushout in the ambient category:

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{j} & Y \\ & & \downarrow d & & \downarrow f \\ & & C & \xrightarrow{k} & E \end{array}$$

By theorem 1.1, we know that f is an epimorphism. It then remains to prove that $j \circ i$ is a kernel of f . Since j is a monomorphism, one can show that the square is also a pullback (see [8, Corollary 13.6] for a proof). Consider $g : X \rightarrow Y$ such that $f \circ g = 0$. Using the pullback property on $g : X \rightarrow Y$ and $0 : X \rightarrow C$, we get a unique map $\hat{g} : X \rightarrow B$ such that $j \circ \hat{g} = g$ and $d \circ \hat{g} = 0$. Now, using that i is a kernel of d , we get a unique map $\hat{\hat{g}} : X \rightarrow A$ such that $i \circ \hat{\hat{g}} = \hat{g}$. In particular, $(j \circ i) \circ \hat{\hat{g}} = g$, and $\hat{\hat{g}}$ is the unique map satisfying this property. Finally,

we note that

$$f \circ (j \circ i) = (f \circ j) \circ i = (k \circ d) \circ i = k \circ (d \circ i) = 0,$$

so we can conclude that $j \circ i$ is a kernel of f . Since f is an epimorphism and $j \circ i$ is its kernel, we conclude that $A \xrightarrow{j \circ i} Y \xrightarrow{f} E$ is a short exact sequence in $\mathcal{C}$. As such, inflations are closed under composition. Proving $[\mathbf{E1}^{\text{op}}]$ is dual. $\square$

Remark 1.12 A slightly modified converse statement is also true, and is known as the *Gabriel-Quillen embedding theorem*. It states that, if $(\mathcal{A}, \mathcal{E})$ is an exact category with $\mathcal{A}$ small, there exists an abelian category $\mathcal{B}$ and a fully faithful exact functor $i : \mathcal{A} \hookrightarrow \mathcal{B}$ that reflects exactness. Furthermore, $\mathcal{A}$ is closed under extension in $\mathcal{B}$.

1.4 Examples

The previous statement is quite useful for finding examples of exact categories that are not abelian categories. Here are a few fundamental general examples.

Examples 1.13

- Let R be a non-noetherian unital ring. Consider the category of finitely generated left R -modules $\text{mod}R$, seen as a subcategory of $\text{Mod}R$. It is well known that this category is closed under extensions. However, it is not abelian. Indeed, consider a non-finitely generated ideal $I \subseteq R$ (which exists since R is non-noetherian). Then the kernel of the quotient map is $R \twoheadrightarrow R/I$ is the ideal I . In that case, both R and R/I are finitely generated R -modules, but I is by definition not finitely generated. Thus $\text{mod}R$ does not contain the kernel of all its maps, and cannot be abelian.
- Let R be any ring. Consider the full subcategory of $\text{mod}R$ consisting of all projectives $\mathcal{P}(R)$. Let $P, P' \in \mathcal{P}(R)$ be two

projective R -modules. It is well known from homological algebra that $\text{Ext}_R^1(P', P) = 0$. But then, considering Ext_R^1 as the Yoneda extension functor, this implies that any short exact sequence $0 \rightarrow P \rightarrow E \rightarrow P' \rightarrow 0$ splits, i.e. $E \cong P \oplus P'$. Since the direct sum of two projectives is projective, this shows that $\mathcal{P}(R)$ is closed under extensions. However, it is not necessarily abelian. For example, in the category $\text{mod}\mathbb{Z}$, the cokernel of the map $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ is $\mathbb{Z}/2\mathbb{Z}$, which is not projective, so $\mathcal{P}(\mathbb{Z})$ is not abelian.

1.5 Reducing the axioms

As was shown by B. Keller in [5, Appendix A], some of the axioms required in the classical definition of exact categories are redundant. We finish this chapter by providing parts of the proof of this result.

Proposition 1.14 *Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphisms. Assume that $\mathcal{E}$ satisfies $[\text{E1}^{\text{op}}]$ and $[\text{E2}]$, as well as*

$[\text{E0}']$ 1_0 is a deflation;

$[\text{E2}^{\text{op}}']$ For any $f : Z' \rightarrow Z$ and deflation $d : Y \twoheadrightarrow Z$, there is a commutative square

$$\begin{array}{ccc} Y' & \xrightarrow{d'} \twoheadrightarrow & Z' \\ f' \downarrow & & \downarrow f \\ Y & \xrightarrow{d} \twoheadrightarrow & Z \end{array}$$

Then $\mathcal{E}$ is an exact structure on $\mathcal{A}$.

Proof We will first prove that $[\text{E2}]$ and $[\text{E2}^{\text{op}}']$, combined with closure under isomorphisms, imply $[\text{E2}^{\text{op}}]$ by explicitly building a pullback.

Consider the commutative diagram from [E2^{op}]:

$$\begin{array}{ccccc} K & \xrightarrow{i'} & Y' & \xrightarrow{d'} & Z' \\ f_1 \downarrow & & f_2 \downarrow & & f_3 \downarrow \\ X & \xrightarrow{i} & Y & \xrightarrow{d} & Z \end{array}$$

Note that the left part of the diagram is obtained by taking inflations associated to d' and d , and that f_1 exists as the map between kernels. Since we have two morphisms starting in K , one of which being an inflation, we can build the pushout of i' along f_1 , which we will name P . Our strategy will be to show that P is the pullback of d along f_3 . To do so, we first construct the morphisms $\hat{d}' : P \rightarrow Z'$ and $\hat{f}_2 : P \rightarrow Y$ by applying the pushout property to the following commutative diagrams:

$$\begin{array}{ccc} \begin{array}{ccccc} K & \xrightarrow{i'} & Y' & & \\ f_1 \downarrow & & \downarrow \hat{f}_1 & \searrow d' & \\ X & \xrightarrow{i'} & P & \xrightarrow{\hat{d}'} & Z' \\ & \searrow 0 & & \nearrow \exists! \hat{d}' & \end{array} & \text{and} & \begin{array}{ccccc} K & \xrightarrow{i'} & Y' & & \\ f_1 \downarrow & & \downarrow \hat{f}_1 & \searrow f_2 & \\ X & \xrightarrow{i'} & P & \xrightarrow{\hat{f}_2} & Y \\ & \searrow i & & \nearrow \exists! \hat{f}_2 & \end{array} \end{array}$$

For P to be a pullback, it is necessary that $d \circ \hat{f}_2 = f_3 \circ \hat{d}'$ (since a pullback gives a commutative square). But this is true as both maps are a solution to the pushout problem

$$\begin{array}{ccccc} K & \xrightarrow{i'} & Y' & \xrightarrow{d'} & Z' \\ f_1 \downarrow & & \downarrow \hat{f}_1 & & \downarrow f_3 \\ X & \xrightarrow{i'} & P & \xrightarrow{\hat{d}'} & Z' \\ \downarrow i & & \searrow d \hat{f}_2 = f_3 \hat{d}' & & \downarrow f_3 \\ Y & \xrightarrow{i} & Y & \xrightarrow{d} & Z \end{array}$$

Clearly $(d \circ i) \circ f_1 = 0 = f_3 \circ (d' \circ i')$, so the big square commutes. Thus

there exists a unique map $p : P \rightarrow Z$ such that $p \circ \hat{f}_1 = f_3 \circ d'$ and $p \circ \hat{i}' = d \circ i = 0$. But note that $f_3 \circ (\hat{d}' \circ \hat{f}_1) = f_3 \circ d'$ and $f_3 \circ (\hat{d}' \circ \hat{i}') = 0$ by definition of $\hat{d}'$, so $f_3 \circ \hat{d}' = p$. Similarly, $d \circ (\hat{f}_2 \circ \hat{f}_1) = d \circ f_2 = f_3 \circ d'$ by commutativity and $d \circ (\hat{f}_2 \circ \hat{i}') = d \circ i = 0$, so $d \circ \hat{f}_2 = p = f_3 \circ \hat{d}'$.

Next, we will show that $\hat{d}'$ is a cokernel of $\hat{i}'$. Consider a morphism $g : P \rightarrow D$ for some $D \in \mathcal{A}$ such that $g \circ \hat{i}' = 0$. Clearly, this implies that $g \circ (\hat{f}_1 \circ i') = g \circ (\hat{i}' \circ f_1) = 0$. Using that d' is a cokernel of i' , there exists a unique morphism $k : Z' \rightarrow D$ such that $k \circ d' = g \circ \hat{f}_1$. What is left to show is that $k \circ \hat{d}' = g$, but this is true since they are both solutions to the pushout problem

$$\begin{array}{ccccc}
 K & \xrightarrow{i'} & Y' & & \\
 f_1 \downarrow & & \hat{f}_1 \downarrow & \searrow & \\
 X & \xrightarrow{\quad} & P & \xrightarrow{\hat{f}_1 g = k d'} & D \\
 & \searrow \hat{i}' & & \nearrow k \hat{d}' = g & \\
 & & 0 & \searrow &
 \end{array}$$

Therefore, there exists a unique map k such that $k \circ \hat{d}' = g$, i.e. $\hat{d}'$ is a cokernel of $\hat{i}'$. This implies the top row in the commutative diagram

$$\begin{array}{ccccc}
 X & \xrightarrow{\hat{i}'} & P & \xrightarrow{\hat{d}'} & Z' \\
 \parallel & & \hat{f}_2 \downarrow & & f_3 \downarrow \\
 X & \xrightarrow{i} & Y & \xrightarrow{d} & Z
 \end{array}$$

is a conflation, which is a requirement in [E2^{op}].

The last step is to prove that the rightmost square is a pullback. To do this, we will need to create a few more morphisms. Consider the

pushout of $\hat{i}'$ along i

$$\begin{array}{ccccc}
 X & \xrightarrow{\hat{i}'} & P & \xrightarrow{\hat{d}'} & Z' \\
 i \downarrow & \swarrow \hat{f}_2 & \downarrow j & & \parallel \\
 Y & \xrightarrow{\hat{j}'} & C & \xrightarrow{\hat{e}'} & Z' \\
 d \downarrow & & \downarrow e & & \\
 Z & = & Z & &
 \end{array}$$

where all rows and columns are conflations due to [E2] (it is not hard to show that one can choose Z' (respectively Z) as the cokernel object of $\hat{j}'$ (respectively j) using the pushout property). Considering the pushout property applied to the commutative diagram

$$\begin{array}{ccc}
 X & \xrightarrow{\hat{i}'} & P \\
 i \downarrow & & \downarrow j \\
 Y & \xrightarrow{\hat{j}'} & C
 \end{array}
 \quad
 \begin{array}{c}
 \searrow \hat{f}_2 \\
 \downarrow \exists! r \\
 Y
 \end{array}$$

we get a unique map $r : C \rightarrow Y$ such that $r \circ \hat{j}' = 1_Y$ and $r \circ j = \hat{f}_2$. Since

$$(1_C - \hat{j}' \circ r) \circ \hat{j}' = \hat{j}' - \hat{j}' \circ (r \circ \hat{j}') = \hat{j}' - \hat{j}' = 0,$$

we get a unique map $g : Z' \rightarrow C$ such that $g \circ \hat{d}' = 1_C - \hat{j}' \circ r$.

Note that $-e \circ g = f_3$, since both $f_3 \circ \hat{e}'$ and $-e \circ g \circ \hat{e}'$ are solutions of the pushout problem

$$\begin{array}{ccc}
 X & \xrightarrow{\hat{i}'} & P \\
 i \downarrow & & \downarrow j \\
 Y & \xrightarrow{\hat{j}'} & C
 \end{array}
 \quad
 \begin{array}{c}
 \searrow f_3 \hat{d}' \\
 \downarrow -e g \hat{e}' = f_3 \hat{e}' \\
 Z
 \end{array}$$

Indeed, that both compositions with $\hat{j}'$ are zero is direct since $(\hat{j}', \hat{e}')$ is a conflation. And since $-e \circ (g \circ \hat{e}') = -e \circ (1_C - \hat{j}' \circ r) = -e + d \circ r$, we get that

$$(-e \circ g \circ \hat{e}') \circ j = (-e + d \circ r) \circ j = -e \circ j + d \circ b = f_3 \circ \hat{d}'.$$

so that it solves the pushout problem. Similarly, $f_3 \circ (\hat{e}' \circ j) = f_3 \circ \hat{d}'$, so it solves the pushout problem as well. Thus, $-e \circ g \circ \hat{e}' = f_3 \circ \hat{e}'$ and therefore $-e \circ g = f_3$ since $\hat{e}'$ is an epimorphism.

Now we are ready to show that P is the pullback of d along f_3 . Consider the commutative diagram

$$\begin{array}{ccccc} D & & & & \\ & \searrow z & & & \\ & & P & \xrightarrow{\hat{d}'} & Z' \\ & & \downarrow \hat{f}_2 & & \downarrow f_3 \\ & & Y & \xrightarrow{d} & Z \\ & \nearrow y & & & \end{array}$$

Note that $f_3 \circ z = d \circ y$ if and only if $e \circ (\hat{j}' \circ y + g \circ z) = 0$ (using that $d = e \circ \hat{j}'$ and $f_3 = -e \circ g$). Thus, using that j is a kernel of e , we get a unique map $h : D \rightarrow P$ such that $j \circ h = \hat{j}' \circ y + g \circ z$. Then, note that $r \circ g \circ \hat{e}' = r \circ (1_C - \hat{j}' \circ r) = r - r = 0$, and therefore $r \circ g = 0$. Thus,

$$\hat{f}_2 \circ h = r \circ j \circ h = r \circ \hat{j}' \circ y + r \circ g \circ z = y.$$

In addition, since $\hat{e}' \circ g \circ \hat{e}' = \hat{e}' \circ (1_C - \hat{j}' \circ r) = \hat{e}'$ as $(\hat{j}', \hat{e}')$ is a conflation, we get that $\hat{e}' \circ g = 1_{Z'}$. Finally,

$$\hat{d}' \circ h = \hat{e}' \circ j \circ h = \hat{e}' \circ \hat{j}' \circ y + \hat{e}' \circ g \circ z = z.$$

Thus, we get that $\hat{d}' \circ h = z$ and $\hat{f}_2 \circ h = y$, i.e. h solves the pullback problem. This allows us to conclude that the square

$$\begin{array}{ccc} P & \xrightarrow{\hat{d}'} & Z' \\ \hat{f}_2 \downarrow & & \downarrow f_3 \\ Y & \xrightarrow{d} & Z \end{array}$$

is indeed a pullback, and shows that $[\text{E2}^{\text{op}}]$ holds.

We will not prove that, if $[\text{E0}']$, $[\text{E1}^{\text{op}}]$, $[\text{E2}]$ and $[\text{E2}^{\text{op}}]$ hold, then so does $[\text{E1}]$, but refer the reader to [5, Appendix A.1].

Thus, it only remains to show that $[\text{E0}']$ implies both $[\text{E0}]$ and $[\text{E0}^{\text{op}}]$. First, note that 1_0 is also an inflation, since it is its own kernel, and it is a deflation. Next, note that the commutative square

$$\begin{array}{ccc} X & \xrightarrow{1_X} & X \\ 0 \downarrow & \lrcorner & \downarrow 0 \\ 0 & \xrightarrow{0} & 0 \end{array}$$

is a pullback. Indeed, let $D \in \mathcal{A}$ fit into the commutative diagram

$$\begin{array}{ccccc} D & & & & \\ & \searrow g & & \searrow & \\ & & X & \xrightarrow{1_X} & X \\ & \searrow 0 & \downarrow 0 & & \downarrow 0 \\ & & 0 & \xrightarrow{0} & 0 \end{array}$$

Then the only morphism that can fit as the dashed arrow is g itself, and the pullback property is trivially fulfilled. Therefore, because of $[\text{E2}^{\text{op}}]$, 1_X is a deflation. Similarly, but using a pushout instead, 1_X is also an inflation. This proves that both $[\text{E0}]$ and $[\text{E0}^{\text{op}}]$ are true. $\square$

2 Generalizing to n -exact categories

After generalizing from abelian categories to exact categories, we can go one step further by looking at longer exact sequences. First, we will redefine the notions needed for exact categories, but in a more general context. Then we will proceed to define n -exact categories, and talk about a few properties. Finally, we will give some concrete examples of such categories.

The content of this chapter is based on [4]. We let $\mathcal{A}$ denote an additive category when not stated otherwise.

2.1 Working in higher degrees

Before defining n -exact categories, there are many prerequisites that need to be addressed. Firstly, we will need a way to properly describe what the analogue of a kernel-cokernel pair would be in this new context. As a result, it makes sense to define n -kernels and n -cokernels, which require the notion of weak kernel (respectively weak cokernel).

Definition 2.1 Let $f : A \rightarrow B$ be a morphism in $\mathcal{A}$. A *weak kernel* of f is an object $K \in \mathcal{A}$ with a morphism $k : K \rightarrow A$ such that the sequence of abelian groups

$$\mathcal{A}(C', K) \xrightarrow{k \circ -} \mathcal{A}(C', A) \xrightarrow{f \circ -} \mathcal{A}(C', B)$$

is exact for all $C' \in \mathcal{A}$.

Dually, a *weak cokernel* of f is an object $C \in \mathcal{A}$ with a morphism $c : B \rightarrow C$ such that the sequence of abelian groups

$$\mathcal{A}(C, C') \xrightarrow{- \circ c} \mathcal{A}(B, C') \xrightarrow{- \circ f} \mathcal{A}(A, C')$$

is exact for all $C' \in \mathcal{A}$.

Remarks 2.2 We do not ask for the sequence to be short exact, but only for it to be exact in the middle term. Furthermore, the following observations are quite useful in practice:

- (a) Notice that the direction of the sequences is different since for weak cokernels, we apply the contravariant functor $\mathcal{A}(-, C')$ to our original sequence, whereas we apply the covariant functor $\mathcal{A}(C', -)$ in the case of weak kernels;
- (b) For $k : K \rightarrow A$ a weak kernel of $f : A \rightarrow B$, the associated Hom-sequence being exact lets us know that
 - (i) The composition $f \circ k$ is 0;
 - (ii) For any morphism $h : C \rightarrow A$ such that $f \circ h = 0$, there exists a morphism $\hat{h} : C \rightarrow K$ such that $k \circ \hat{h} = h$.

That last point can be summarized by the following diagram:

$$\begin{array}{ccccc}
 K & \xrightarrow{k} & A & \xrightarrow{f} & B \\
 \uparrow \exists \hat{h} & \nearrow h & \searrow 0 & & \\
 C & & & &
 \end{array}$$

These two points follow from exactness in the middle term. Indeed, since $\text{Im}(k \circ -) = \text{Ker}(f \circ -)$, any map $h : C \rightarrow A$ such that $f \circ h = 0$, is in the image of the post-composition with k , i.e. can be factored (not necessarily uniquely!) as $k \circ \hat{h}$. In particular, if we consider $C = K$, we conclude that

$$k = k \circ 1_K \in \text{Im}(k \circ -) = \text{Ker}(f \circ -),$$

i.e. $f \circ k = 0$. The result for weak cokernels is dual.

In summary, weak cokernels and weak kernels have a very similar property to that of cokernels and kernels, except that uniqueness in our factoring of morphisms is not guaranteed. In other words, a weak kernel is not necessarily a monomorphism, and a weak cokernel is not necessarily an epimorphism.

This notion is quite fundamental in the elaboration of n -kernels and n -cokernels.

Definition 2.3 Let $d^0 : X^0 \rightarrow X^1$ be a morphism in $\mathcal{A}$. An n -kernel of d^0 is a sequence

$$X^{-n} \xrightarrow{d^{-n}} \cdots \xrightarrow{d^{-1}} X^0 \left(\xrightarrow{d^0} X^1 \right)$$

such that the sequence

$$0 \rightarrow \mathcal{A}(C', X^{-n}) \xrightarrow{d^{-n} \circ -} \mathcal{A}(C', X^{-n+1}) \xrightarrow{d^{-n+1} \circ -} \cdots \xrightarrow{d^0 \circ -} \mathcal{A}(C', X^1)$$

induced by applying the functor $\mathcal{A}(C', -)$ is exact.

Equivalently, $(d^{-n}, d^{-n+1}, \dots, d^{-1})$ is an n -kernel of d^0 if and only if d^{-k} is a weak cokernel of d^{-k+1} for all $1 \leq k \leq n-1$, and d^{-n} is a kernel of d^{-n+1} .

Dually, an n -cokernel of d^0 is a sequence

$$\left(X^0 \xrightarrow{d^0} \right) X^1 \xrightarrow{d^1} X^2 \xrightarrow{d^2} \cdots \xrightarrow{d^n} X^{n+1}$$

such that d^k is a weak cokernel of d^{k-1} for $1 \leq k \leq n-1$ and d^n is a cokernel of d^{n-1} .

Remark 2.4 In the case $n = 1$, this indeed coincides with the usual definition of kernels and cokernels.

Now that we have a suitable notion of higher kernels and cokernels, we can consider the concept of n -exact sequences.

Definition 2.5 An n -exact sequence in $\mathcal{A}$ is a chain complex

$$X^0 \xrightarrow{d^0} X^1 \xrightarrow{d^1} \dots \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1}$$

in $\text{Ch}^{n+1}(\mathcal{A})$ such that $(d^0, \dots, d^{n-1})$ is an n -kernel of d^n and $(d^1, \dots, d^n)$ is an n -cokernel of d^0 .

Remark 2.6 Once again, when $n = 1$, we recover the usual notion of a short exact sequence.

There are three final notions that need to be introduced before proceeding with the definition of n -exact categories: that of n -pullback, n -pushout, and weak isomorphisms. Before covering these concepts, let us first recall what the cone of a morphism of complexes is:

Definition 2.7 Let X and Y be two complexes in $\text{Ch}(\mathcal{A})$ and $f : X \rightarrow Y$ be a morphism of complexes, or diagrammatically:

$$\begin{array}{ccccccc} X : & \dots & \longrightarrow & X^{k-1} & \xrightarrow{d_X^{k-1}} & X^k & \xrightarrow{d_X^k} & X^{k+1} & \longrightarrow & \dots \\ & & & \downarrow f^{k-1} & & \downarrow f^k & & \downarrow f^{k+1} & & \\ Y : & \dots & \longrightarrow & Y^{k-1} & \xrightarrow{d_Y^{k-1}} & Y^k & \xrightarrow{d_Y^k} & Y^{k+1} & \longrightarrow & \dots \end{array}$$

The *cone of f* , denoted $C(f)$, is the complex

$$C(f) : \dots \xrightarrow{d_C^{k-2}} X^k \oplus Y^{k-1} \xrightarrow{d_C^{k-1}} X^{k+1} \oplus Y^k \xrightarrow{d_C^k} \dots$$

where $d_C^k = \begin{bmatrix} -d_X^{k+1} & 0 \\ f^{k+1} & d_Y^k \end{bmatrix}$.

Before looking at the higher analogues of pullbacks and pushouts, it is quite useful to examine the link that they have with the cone of a morphism.

Remark 2.8 Let $B^1 \xrightarrow{d_B^1} B^2$ and $A^2 \xrightarrow{f^2} B^2$ be two morphisms in $\mathcal{A}$ and consider the pullback of d_B^1 along f^2 :

$$\begin{array}{ccc} A^1 & \xrightarrow{d_A^1} & A^2 \\ f^1 \downarrow & \lrcorner & \downarrow f^2 \\ B^1 & \xrightarrow{d_B^1} & B^2 \end{array}$$

Clearly, $f = (f^1, f^2)$ is a morphism of complexes since the square commutes. Thus, we can consider its cone:

$$C(f) : \quad 0 \longrightarrow A^1 \xrightarrow{d_C^0} A^2 \oplus B^1 \xrightarrow{d_C^1} B^2 \longrightarrow 0$$

where $d_C^0 = \begin{bmatrix} -d_A^1 \\ f^1 \end{bmatrix}$, and $d_C^1 = \begin{bmatrix} f^2 & d_B^1 \end{bmatrix}$. We claim that d_C^0 is a kernel of d_C^1 . Indeed, consider a morphism $\begin{bmatrix} g \\ h \end{bmatrix} : C \rightarrow A^2 \oplus B^1$ such that the composition $d_C^1 \circ \begin{bmatrix} g \\ h \end{bmatrix} = 0$. The composition being 0 implies that $f^2 \circ g + d_B^1 \circ h = 0$, or $d_B^1 \circ h = -f^2 \circ g$. Consider the unique map $\hat{h} : C \rightarrow A^1$ given by $f^1 \circ \hat{h} = h$ and $d_A^1 \circ \hat{h} = -g$ (this map comes from the pullback property). We see that $d_C^0 \circ \hat{h} = \begin{bmatrix} -d_A^1 \circ \hat{h} \\ f^1 \circ \hat{h} \end{bmatrix} = \begin{bmatrix} g \\ h \end{bmatrix}$. Thus we have defined a unique map $\hat{h} : C \rightarrow A^1$ such that $d_C^0 \circ \hat{h} = \begin{bmatrix} g \\ h \end{bmatrix}$. That is, d_C^0 is a kernel of d_C^1 .

A dual result is true for pushouts, where d_C^1 is then a cokernel of d_C^0 .

Basing ourselves on this observation, we can generalize the notion of pullback and pushout to higher degrees.

Definition 2.9 Let $X : X^0 \xrightarrow{d_X^0} X^1 \xrightarrow{d_X^1} \dots \xrightarrow{d_X^{n-1}} X^n$ be a complex in $\text{Ch}^n(\mathcal{A})$ and $f^0 : X^0 \rightarrow Y^0$ a morphism in $\mathcal{A}$. An n -pushout of X along f^0 is a morphism of complexes

$$\begin{array}{ccccccc} X : & X^0 & \xrightarrow{d_X^0} & X^1 & \xrightarrow{d_X^1} & \dots & \xrightarrow{d_X^{n-1}} X^n \\ & \downarrow f^0 & & \downarrow f^1 & & & \downarrow f^n \\ Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \dots & \xrightarrow{d_Y^{n-1}} Y^n \end{array}$$

such that in the cone of f

$$C(f) : \quad X^0 \xrightarrow{d_C^{-1}} X^1 \oplus Y^0 \xrightarrow{d_C^0} \dots \xrightarrow{d_C^{n-1}} X^n \oplus Y^{n-1} \xrightarrow{d_C^n} Y^n$$

we have that $(d_C^0, \dots, d_C^n)$ is an n -cokernel of d_C^{-1} .

The definition for the n -pullback of a chain complex is dual. That is, given a complex Y in $\text{Ch}^n(\mathcal{A})$ and a morphism $f^n : X^n \rightarrow Y^n$, an n -pullback of Y along f^n is a morphism of complexes $f : X \rightarrow Y$ such that in its cone $C = C(f)$, the sequence $(d_C^{-1}, \dots, d_C^{n-1})$ is an n -kernel of d_C^n .

Remark 2.10 When $n = 1$, we recover the classical notion of pullbacks and pushouts.

The last notion required for defining n -exact categories is that of weak isomorphisms:

Definition 2.11 Let X and Y be two n -exact sequences in $\mathcal{A}$. We say that $f : X \rightarrow Y$ is a *weak isomorphism* of complexes if there is $k \in \{0, 1, \dots, n, n+1\}$ such that f^k and f^{k+1} are isomorphisms, with the convention that $n+2 := 0$.

When $n = 1$, we note that weak isomorphisms are isomorphisms of short exact sequences. This can be shown by using the five lemma, which holds for any exact category (see [3, Lemma 8.9]). In the general

case, however, weak isomorphisms are not necessarily isomorphisms, but a weaker result still holds.

Proposition 2.12 *Let X and Y be two n -exact sequences in $\mathcal{A}$ and let $f : X \rightarrow Y$ be a morphism of complexes. If f is a weak isomorphism, then f induces an isomorphism in $H(\mathcal{A})$.*

A proof of this result can be found in [4, Proposition 2.7].

All of this preliminary work being done, we can finally proceed to define n -exact categories.

2.2 n -exact categories

With the proper tools put in place, it is quite straightforward to generalize the definition of exact categories to longer exact sequences. Similarly as in Chapter 1, we proceed with admissible n -exact sequences, admissible monomorphisms, and admissible epimorphisms.

Definition 2.13 Let $\mathcal{E}$ be a class of n -exact sequences in $\mathcal{A}$. An $\mathcal{E}$ -admissible n -exact sequence is an n -exact sequence that is in $\mathcal{E}$. An $\mathcal{E}$ -admissible monomorphism is a morphism $d_X^0 : X^0 \rightarrow X^1$ such that there is an $\mathcal{E}$ -admissible n -exact sequence

$$X^0 \xrightarrow{d_X^0} X^1 \xrightarrow{d_X^1} \dots \xrightarrow{d_X^n} X^{n+1}.$$

Dually, an $\mathcal{E}$ -admissible epimorphism is a morphism $d_X^n : X^n \rightarrow X^{n+1}$ such that there is an $\mathcal{E}$ -admissible n -exact sequence

$$X^0 \xrightarrow{d_X^0} \dots \xrightarrow{d_X^{n-1}} X^n \xrightarrow{d_X^n} X^{n+1}.$$

Whenever the context is clear, we will simply say admissible n -exact sequence (respectively admissible monomorphism, admissible epimor-

phism). We will write $X^0 \xrightarrow{d_X^0} X^1$ for admissible monomorphisms and $X^n \xrightarrow{d_X^n} X^{n+1}$ for admissible epimorphisms.

The following definition adapts the axioms of exact categories to the definitions given in the previous chapter.

Definition 2.14 Let $\mathcal{E}$ be a class of n -exact sequences on $\mathcal{A}$. $\mathcal{E}$ is said to be an n -exact structure on $\mathcal{A}$ if it is closed under weak isomorphisms and it satisfies the following axioms:

- [E0] The sequence $0 \rightarrow 0 \rightarrow \cdots \rightarrow 0$ is an $\mathcal{E}$ -admissible n -exact sequence;
- [E1] The composition of two $\mathcal{E}$ -admissible monomorphisms yields an $\mathcal{E}$ -admissible monomorphism;
- [E1^{op}] The composition of two $\mathcal{E}$ -admissible epimorphisms yields an $\mathcal{E}$ -admissible epimorphism;
- [E2] If $X : X^0 \xrightarrow{d_X^0} \cdots \xrightarrow{d_X^n} X^{n+1}$ is an $\mathcal{E}$ -admissible n -exact sequence and $X^0 \xrightarrow{f^0} Y^0$ is a morphism in $\mathcal{A}$, then there exists an n -pushout

$$\begin{array}{ccccccc}
 X : & X^0 & \xrightarrow{d_X^0} & X^1 & \xrightarrow{d_X^1} & \cdots & \xrightarrow{d_X^{n-1}} X^n \cdots \cdots \twoheadrightarrow X^{n+1} \\
 \downarrow f & \downarrow f^0 & & \downarrow f^1 & & & \downarrow f^n \\
 Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \cdots & \xrightarrow{d_Y^{n-1}} Y^n
 \end{array}$$

of $(d_X^0, \dots, d_X^{n-1})$ along f^0 such that d_Y^0 is an $\mathcal{E}$ -admissible monomorphism;

- [E2^{op}] If $Y : Y^0 \xrightarrow{d_Y^0} \cdots \xrightarrow{d_Y^n} Y^{n+1}$ is an $\mathcal{E}$ -admissible n -exact sequence and $X^{n+1} \xrightarrow{f^{n+1}} Y^{n+1}$ is a morphism in $\mathcal{A}$, then there is an n -pullback

$$\begin{array}{ccccccc}
 X : & & & X^1 & \xrightarrow{d_X^1} & \cdots & \xrightarrow{d_X^{n-1}} X^n \xrightarrow{d_X^n} \twoheadrightarrow X^{n+1} \\
 \downarrow f & & & \downarrow f^1 & & & \downarrow f^n \\
 Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \cdots & \xrightarrow{d_Y^{n-1}} Y^n \xrightarrow{d_Y^n} \twoheadrightarrow Y^{n+1}
 \end{array}$$

of $(d_Y^1, \dots, d_Y^n)$ along f^{n+1} such that d_X^n is an $\mathcal{E}$ -admissible epimorphism.

If $\mathcal{E}$ is an n -exact structure on $\mathcal{A}$, we say that $(\mathcal{A}, \mathcal{E})$ is an *n -exact category*.

Remarks 2.15 • The axiom $[E0]$ given here seems to be less restrictive than the $[E0]$ and $[E0^{op}]$ of exact categories. However, recall that we showed in theorem 1.14 that a similar axiom is also sufficient in the exact case. Furthermore, if $n \geq 2$, note that identity morphisms are also admissible monomorphisms since

$$\begin{array}{ccccccc} X & \xrightarrow{1_X} & X & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & & & \downarrow \\ 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 \end{array}$$

is a weak isomorphism. Identity morphisms are admissible epimorphisms for the same reason.

- One can show that n -kernels and n -cokernels are unique up to weak isomorphism. As such, all n -kernels (respectively n -cokernels) of an admissible epimorphism (respectively admissible monomorphism) fit into an admissible n -exact sequence.

Many classical results from exact categories can be generalized to n -exact categories without much trouble. For example, there is a way to generalize Quillen's obscure axiom.

Proposition 2.16 (Obscure axiom) *Let $(\mathcal{A}, \mathcal{E})$ be an n -exact category. Suppose there is a commutative diagram*

$$\begin{array}{ccccccc} X : & & X^0 & \xrightarrow{d_X^0} & \dots & \xrightarrow{d_X^{n-1}} & X^n \xrightarrow{d_X^n} X^{n+1} \\ & \downarrow f & \downarrow f^0 & & & \downarrow f^n & \parallel \\ Y : & & Y^0 & \xrightarrow{d_Y^0} & \dots & \xrightarrow{d_Y^{n-1}} & Y^n \xrightarrow{d_Y^n} Y^{n+1} \end{array}$$

where the top row is an $\mathcal{E}$ -admissible n -exact sequence and $(d_Y^0, \dots, d_Y^{n-1})$ is an n -kernel of d_Y^n . Then Y is an $\mathcal{E}$ -admissible n -exact sequence.

A proof of the dual statement can be found in [4, Proposition 4.11].

Remark 2.17 What is really required for the statement to hold is a morphism $f^n : X^n \rightarrow Y^n$ such that $d_X^n = d_Y^n f^n$. Indeed, such a map can be extended to a morphism between n -kernels in much the same way that it induces a morphism between kernels in the exact case. Since $d_Y^n(f^n d_X^{n-1}) = (d_Y^n f^n) d_X^{n-1} = d_X^n d_X^{n-1} = 0$, there exists a (not necessarily unique) morphism $f^{n-1} : X^{n-1} \rightarrow Y^{n-1}$ such that $d_Y^{n-1} f^{n-1} = f^n d_X^{n-1}$, since d_Y^{n-1} is a weak kernel of d_Y^n . Playing the same game again with f^{n-1} , we can create a morphism of complexes $f : X \rightarrow Y$, giving us the required setup.

A convenient property of n -exact categories is that admissible n -exact sequences interact nicely with direct sums. This gives an easy way to construct new n -exact sequences.

Proposition 2.18 *Let $(\mathcal{A}, \mathcal{E})$ be an n -exact category and $X, Y \in \text{Ch}^{n+1}(\mathcal{A})$. Then X and Y are admissible n -exact sequences if and only if $X \oplus Y$ is an admissible n -exact sequence.*

Proof The proof is inspired by those of [4, Proposition 4.6] and [4, Proposition 4.12].

Assume first that both X and Y are admissible n -exact sequences. Consider the sequence

$$X^0 \oplus Y^0 \xrightarrow{d_X^0 \oplus 1_{Y^0}} X^1 \oplus Y^0 \xrightarrow{\begin{bmatrix} d_X^1 & 0 \end{bmatrix}} X^2 \xrightarrow{d_X^2} \dots \xrightarrow{d_X^{n-1}} X^n \xrightarrow{d_X^n} X^{n+1}.$$

Since it differs from X only by a contractible complex, it is weakly isomorphic to X , and therefore an admissible n -exact sequences. Thus $d_X^0 \oplus 1_{Y^0}$ is an admissible monomorphism. Similarly, $1_{X^1} \oplus d_Y^0$ is an admissible monomorphism. Thus, because of [E1],

$$d_X^0 \oplus d_Y^0 = (1_{X^1} \oplus d_Y^0) \circ (d_X^0 \oplus 1_{Y^0})$$

is an admissible monomorphism, and $(d_X^1 \oplus d_Y^1, \dots, d_X^n \oplus d_Y^n)$ is one of its n -cokernels. Hence, $X \oplus Y$ is a admissible n -exact sequence.

Conversely, assume $X \oplus Y$ is an admissible n -exact sequence. Then, we can exhibit $\begin{bmatrix} d_X^n & 0 \end{bmatrix} = \begin{bmatrix} 1_{X^{n+1}} & 0 \end{bmatrix} \circ (d_X^n \oplus d_Y^n)$ (which is an admissible epimorphism by [E1^{op}]) as

$$\begin{array}{ccc} X^n \oplus Y^n & \xrightarrow{d_X^n \oplus d_Y^n} & X^{n+1} \oplus Y^{n+1} \xrightarrow{\begin{bmatrix} 1_{X^{n+1}} & 0 \end{bmatrix}} X^{n+1} \\ \begin{bmatrix} 1_X^n & 0 \end{bmatrix} \downarrow & & \parallel \\ X^n & \xrightarrow{d_X^n} & X^{n+1} \end{array}$$

Thus d_X^n is an admissible epimorphism by theorem 2.16, and admits as n -kernel $(d_X^0, \dots, d_X^{n-1})$, i.e. X is an admissible n -exact sequence. The same can be said about Y . $\square$

2.3 n -cluster tilting subcategories

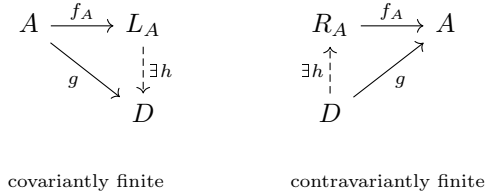
Similar to extension-closed subcategories of abelian categories, there is a higher degree notion that gives us natural examples of n -exact

categories. However, this requires a few more definitions.

Definition 2.20 Let $\mathcal{A}$ be an additive category and $\mathcal{D}$ be a full subcategory of $\mathcal{A}$. $\mathcal{D}$ is called *covariantly finite* if for any object $A \in \mathcal{A}$ there exists an object $L_A \in \mathcal{D}$ and a morphism $f_A : A \rightarrow L_A$ such that for any morphism $g : A \rightarrow D$ where $D \in \mathcal{D}$, there exists a morphism $h : L_A \rightarrow D$ such that $g = h \circ f_A$.

Dually, $\mathcal{D}$ is called *contravariantly finite* if for any object $A \in \mathcal{A}$ there exists an object $R_A \in \mathcal{D}$ and a morphism $f_A : R_A \rightarrow A$ such that for any morphism $g : D \rightarrow A$ where $D \in \mathcal{D}$, there exists a morphism $h : D \rightarrow R_A$ such that $g = f_A \circ h$.

Both properties can be summarized in the following diagrams:



The subcategory $\mathcal{D}$ is said to be *functorially finite* if it is both covariantly and contravariantly finite.

Having introduced the notion of functorial finiteness, there is one last definition required to discuss n -cluster tilting subcategories in a general context, that of generating-cogenerating subcategories.

Definition 2.21 Let $\mathcal{A}$ be an additive category and $\mathcal{D}$ be a full additive subcategory of $\mathcal{A}$. The subcategory $\mathcal{D}$ is said to be *generating* if, for any object $A \in \mathcal{A}$, there exists an object $G_A \in \mathcal{D}$ and an epimorphism $q : G_A \rightarrow A$. Dually, the subcategory $\mathcal{D}$ is called *cogenerating* if, for any object $A \in \mathcal{A}$, there exists an object $C_A \in \mathcal{D}$ and an epimorphism $i : A \rightarrow C_A$.

Remark 2.22 In the context of module categories, theorem 2.21 is always satisfied for any subcategory, since module categories have enough

injectives and projectives. Furthermore, theorem 2.20 is sometimes trivially satisfied for any full additive subcategory. In fact, if A is an Artin algebra, one can show that every subcategory of $\text{mod} A$ is functorially finite if and only if A is of finite representation type (see [2, Proposition 1.2]).

By adding some criteria on how the Ext functor acts with the subcategory, we come to the definition of n -cluster tilting subcategories.

Definition 2.23 Let $\mathcal{A}$ be an abelian category and $\mathcal{D}$ be a full generating-cogenerating subcategory of $\mathcal{A}$. The subcategory $\mathcal{D}$ is an *n -cluster tilting subcategory* of $\mathcal{A}$ if

- it is functorially finite;
- $\mathcal{D} = \{C \in \mathcal{A} \mid \text{Ext}_{\mathcal{A}}^i(C, D) = 0 \ \forall D \in \mathcal{D} \ \forall i \in \{1, \dots, n-1\}\};$
- $\mathcal{D} = \{C \in \mathcal{A} \mid \text{Ext}_{\mathcal{A}}^i(D, C) = 0 \ \forall D \in \mathcal{D} \ \forall i \in \{1, \dots, n-1\}\}.$

Finding n -cluster tilting categories is a good way to create examples of n -exact categories, as was proven in [4, Theorem 3.16].

Theorem 2.24 *Let $\mathcal{A}$ be an abelian category and $\mathcal{D}$ be an n -cluster tilting subcategory of $\mathcal{A}$. Then $\mathcal{D}$ is an n -exact category with the n -exact structure given by the class of all n -exact sequences in $\mathcal{D}$.*

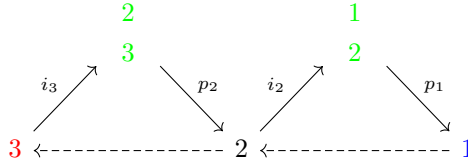
Remarks 2.25

- Jasso actually proved a stronger result, stating that $\mathcal{D}$ is even an n -abelian category. However, we will not go further into this, and only refer to [4, Section 3.1];
- For n -abelian categories, a slightly modified converse statement also holds; see [6, Corollary 1.3 (ii)];
- A similar theory can be built for exact categories instead of abelian categories, which leads to obtaining n -exact categories that are not necessarily n -abelian; see [4, Section 4.2]. Again, a converse statement of sorts also exists; see [7, Theorem 5.3].

2.4 Building concrete examples from AR-quivers of path algebras

When working in sufficiently nice module categories, it is quite easy to check whether a subcategory is n -cluster tilting by using the AR-quiver and interpreting the Ext_Λ^i -functor as i -extensions.

Example 2.26 Consider $\Lambda = k(1 \xrightarrow{\alpha} 2 \xrightarrow{\beta} 3) / \langle \beta\alpha \rangle$. Its AR-quiver is simply



The green modules are indecomposable projective-injectives (from left to right $P_2 = I_3$ and $P_1 = I_2$), the red one is an indecomposable projective (P_3) and the blue one is an indecomposable injective (I_1).

Consider the full subcategory $\mathcal{M}$ of $\text{mod } \Lambda$ given by the additive closure of the colored indecomposables, i.e. $M \in \mathcal{M}$ if and only if $M \cong P_3^{r_1} \oplus P_2^{r_2} \oplus P_1^{r_3} \oplus I_1^{r_4}$ for some integers $r_i \geq 0$.

First, the subcategory $\mathcal{M}$ is functorially finite since Λ is an Artin algebra of finite representation type. Let us next show that $\mathcal{M}$ is 2-cluster tilting. To do so, one must show that both $\text{Ext}_\Lambda^1(M, \mathcal{M}) = 0$ and $\text{Ext}_\Lambda^1(\mathcal{M}, M) = 0$ for any $M \in \mathcal{M}$, and that both are non-zero if $M \notin \mathcal{M}$. Note that, since Ext_Λ^1 is an additive functor, it is enough to check this for indecomposables. For the former, note that the only possible 3-term sequences made of indecomposable modules with end terms in $\mathcal{M}$ and non-zero maps are $P_3 \xrightarrow{i_3} P_2 \xrightarrow{i_2 p_2} P_1$ and $P_2 \xrightarrow{i_2 p_2} P_1 \xrightarrow{p_2} I_1$. However, these are certainly not exact, if only for the fact that $l(P_3) + l(P_1) = 3 \neq 2 = l(P_2)$ and $l(P_2) + l(P_1) = 3 \neq 2 = l(P_1)$.

As for the latter, the only indecomposable candidate that is not in $\mathcal{M}$ is S_2 . Moreover, there is a short exact sequence starting in S_2 (resp. ending in S_2), namely $S_2 \xrightarrow{i_2} P_1 \xrightarrow{p_1} I_1$ (resp. $P_3 \xrightarrow{i_3} P_2 \xrightarrow{p_2} S_2$). But both of these sequences are almost split, and consequently not split. So they are non-zero as elements of $\text{Ext}_\Lambda^1(I_1, S_2)$ (resp. $\text{Ext}_\Lambda^1(S_2, P_3)$). Consequently, we have shown that $\text{Ext}_\Lambda^1(\mathcal{M}, M) \neq 0$ and $\text{Ext}_\Lambda^1(M, \mathcal{M}) \neq 0$ for any module $M \notin \mathcal{M}$.

Having demonstrated that $\mathcal{M}$ is 2-cluster tilting, this gives us the first example of a 2-exact category, with the simplest 2-exact sequence given by

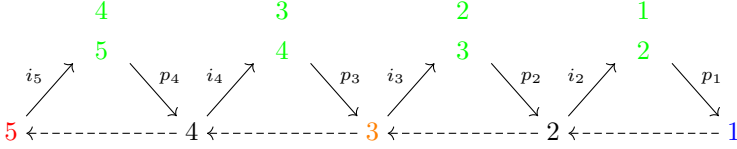
$$P_3 \xrightarrow{i_3} P_2 \xrightarrow{i_2 p_2} P_1 \xrightarrow{p_1} I_1.$$

To see that this sequence is 2-exact, one must check that $(i_3, i_2 p_2)$ is a 2-kernel of p_1 , and that $(i_2 p_2, p_1)$ is a 2-cokernel of i_3 . We will check the first statement, the second one being dual. Consider a map $f : C \rightarrow P_1$ such that $p_1 f = 0$ and $C \in \mathcal{M}$ is an indecomposable module. Then its image is either isomorphic to S_2 , or 0. If its image is 0, one can simply take $\hat{f} : C \rightarrow P_2$ to be the zero-map. If its image is S_2 , then $C = P_2$ is, since this is the only indecomposable module in $\mathcal{M}$ with a non-zero map to S_2 . Thus, one can simply take $\hat{f} = 1_{P_2}$.

Next, consider a map $f : C \rightarrow P_2$ from an indecomposable module $C \in \text{mod } \Lambda$ such that $i_2 p_2 f = 0$. Since i_2 is mono, $p_2 f = 0$. But then since i_3 is a kernel of p_2 , there exists a unique map $\hat{f} : C \rightarrow P_3$ such that $i_3 \hat{f} = f$. This shows that $(i_3, i_2 p_2)$ is a 2-kernel of p_1 .

More generally, if one takes $\Lambda = k(1 \xrightarrow{\alpha_1} \dots \xrightarrow{\alpha_n} n+1) / \langle \{\alpha_{i+1} \alpha_i \mid i \in \{1, \dots, n-1\}\} \rangle$, we get a natural n -tilting subcategory of $\text{mod } \Lambda$, and therefore an example of an n -exact category. However, we may also get shorter exact sequences, leading to lower-degree exact structures as well.

Consider $\Lambda = k(1 \xrightarrow{\alpha_1} \dots \xrightarrow{\alpha_4} 5) / \langle \{\alpha_{i+1}\alpha_i \mid i \in \{1, \dots, 4\}\} \rangle$. The AR-quiver of this algebra is given by



The color code is the same as before, S_3 is colored orange since we will also consider it in our subcategory.

Let $\mathcal{M} \subseteq \text{mod } \Lambda$ be the full subcategory consisting of the additive closure of the colored indecomposable modules. Similarly to what was done before, we can prove that this is a 2-cluster tilting subcategory. This time, there are two natural 2-exact sequences: $X : P_5 \xrightarrow{i_5} P_4 \xrightarrow{i_4 p_4} P_3 \xrightarrow{p_3} S_3$ and $Y : S_3 \xrightarrow{i_3} P_2 \xrightarrow{i_2 p_2} P_1 \xrightarrow{p_1} S_1$. By theorem 2.18, we also get for example that $X \oplus Y$ is 2-exact.

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